American Board of Toxicology Sample Examination

Key for the following questions:
Choose the single most appropriate answer

(1) Which of the following causes functional alterations in monkeys and man that closely resemble the signs and symptoms of Parkinsonism?
A) lead
B) mercury
C) manganese
D) thallium
E) tellurium

(2) A 16-month-old child ingested an unknown quantity of pills which had been prescribed for his mother. Several hours later he was brought to the emergency room because he could not be aroused. On arrival the child was semicomatose and had labored respiration. The pulse was 160, respiration 30, temperature 38.6° C and blood pressure was 80/50. Gavage produced bright red blood and there was blood in the stool. From these symptoms it is likely that the prescription was
A) headache pills
B) birth control pills
C) iron pills
D) diet pills
E) sleeping pills

(3) The clinical presentation associated with the ingestion of kerosene is usually
A) depression of the central nervous system
B) damage of the liver
C) hydrocarbon pneumonitis
D) ventricular arrhythmia
E) renal damage

(4) Which of the following is the major neurotoxic metabolite of n-hexane?
A) 2-butanone (methyl ethyl ketone)
B) 2,5-hexanedione
C) pentane
D) 2,3 hexanedione
E) none of the above

(5) The most commonly used species in screening for delayed neurotoxic effects of organophosphates is
A) dog
B) chicken
C) cat
D) mouse
E) pig

(6) The pesticide DDT exerts its toxic effects on the nervous system primarily by which of the following mechanisms?
A) blocking acetylcholinesterase at cholinergic synapses
B) inhibiting the release of GABA from nerve endings
C) blocking the efflux of calcium from nerve axons
D) blocking postsynaptic acetylcholine receptors
E) delaying the inactivation of sodium channels

(7) All but one of the following atmospheric pollutants is matched with its probable environmental effect. Which agent is matched to an unlikely environmental consequence?
A) chlorofluorocarbons - stratospheric ozone depletion
B) carbon monoxide - spruce forest die-off
C) nitrogen oxides - photochemical smog
D) carbon dioxide - global warming
E) sulfur dioxide - acid precipitation

(8) Chlorine as a gas causes
A) simple asphyxiation
B) methemoglobinemia
C) carboxyhemoglobinemia
D) cholestasis
E) pulmonary edema

(9) A glass bottle containing one curie of (^{3}H) cholesterol is left on the table in the center of a laboratory of 1000 square feet in which several people work. What would most likely be the effect on the workers after two weeks?
A) no effect
B) moderate depression of the lymphocyte count with increased incidence of infections
C) moderate gastrointestinal upset by the end of the two-week period
D) death of at least half of the workers due to acute radiation sickness
E) an increased incidence of malignant tumors after a latent period of about 10 years

Key for the following questions:
Choose the single most appropriate answer

(10) The critical organ for exposure to strontium 90 (beta-radiation) is
A) spleen
B) intestine
C) kidney
D) liver
E) bone ✓

like CALCIUM

(11) The major mode of energy transfer leading to biological damage from ionizing radiation is
A) spallation (breaking apart) of ^{12}C nuclei
B) excitation of DNA atomic electrons
C) electron-positron pair production
D) electron capture
E) ion pair production ✓

NOT p 775

* **(12)** Ingestion of which of the following produces nausea, painful colic and watery diarrhea followed by a symptom-free period of three to five days after which hepatorenal toxicity becomes evident?
A) water hemlock (*Cicuta*)
B) jimsonweed (*Datura stramonium*)
C) oleander (*Nerium oleander*)
D) *Amanita phalloides* ✓
E) *Philodendron*

3-5 DAYS mushroom

(13) A mycotoxin that cause widespread hemorrhage, anemia, granulocytopenia and increased susceptibility to infection in animals is
A) zearalenone
B) T-2 toxin ✓
C) citrinin
D) aflatoxin
E) fumonisin

p39

* **(14)** Which of the following is NOT produced by botulinum toxin?
A) paralysis of skeletal muscles
B) involvement of the third cranial nerve with resultant diplopia and mydriasis
C) dysphagia – swallowing
D) blockade of acetylcholine synthesis ✓
E) blockade of transmission of nerve impulses to the motor end plate

blocks release

* **(15)** Animal carcinogens permitted in food and correct reason
A) saccharin- because of congressional intervention which temporarily abrogates the delaney amendment
B) FD&C red #2 - because the animal study implicating it as a carcinogen was inconclusive
C) aflatoxin - because it is naturally-occurring
D) A and C ✓
E) all of the above

(16) Which of the following is/are true about selenium?
A) dietary deficiencies in young calves result in a fatal disease known as white muscle disease
B) controls lipid peroxidation in conjunction with vitamin E
C) epidemiological data have indicated a decrease in human cancer death rate (age and sex adjusted) that correlates with selenium content of crops
D) there is evidence indicating that selenium has an antineoplastic effect in laboratory animal studies
E) all of the above ✓

(17) Toxic mycotoxins associated with food contamination include which of the following?
A) saxitoxin – shelf-fish – red tide
B) zearalenones ✓
C) saffrole
D) tetrodotoxin
E) none of the above are mycotoxins

(18) Ingestion of tyramine-rich foods such as aged cheese, red wine or beer may precipitate a hypertensive crisis in individuals concurrently taking
A) calcium channel blockers
B) monoamine oxidase inhibitors ✓ – depression
C) sympathomimetic amines
D) phenothiazine
E) quinidine

(19) Characteristic signs of acute morphine poisoning include
A) diarrhea
B) pinpoint pupils
C) depressed respiration
D) A, B and C
E) B and C ✓

Key for the following questions:
Choose the single most appropriate answer

(20) The mechanism by which a sperm contains both members of a chromosomal pair is
A) nondisjunction
B) clastogenesis
C) mitosis
D) mutagenesis
E) apoptosis

(21) Which of the following would be appropriate to use as a positive control for S9 metabolic activation in *a Salmonella typhimurium* (Ames) assay?
A) sodium azide
B) mitomycin C
C) nitrogen mustard
D) 2-aminoanthracene
E) all of the above

(22) The increase in teratogenicity of salicylate when coadministered with sodium benzoate in the rat is directly related to
A) increased fetal blood volume
B) decreased metabolism of salicylate
C) decreased plasma protein binding of salicylate
D) increased maternal blood volume
E) increased excretion of water-soluble metabolites

(23) A majority of congenital malformations in humans is due to
A) drugs
B) emotional stress
C) environmental pollutants
D) infectious agents
E) unknown causes

(24) Many inflammatory mediators in the lung are released from cells participating in the inflammatory process. Which of the following mediators is NOT released by the macrophage/ monocyte?
A) complement factors
B) interleukin-1
C) toxic oxygen species
D) histamine
E) tumor necrosis factor (TNF)

(25) Which of the following diseases of the lung is NOT properly paired with its cause?
A) farmer's lung caused by spores of thermophilic actinomyces
B) bagassosis caused by dust from dried sugar cane
C) byssinosis caused by cotton flax or hemp dust
D) asthma-like symptoms caused by toluene diisocyanate
E) cheese washer's lung caused by exposure to toxin eaten by cattle and concentrated in their milk

(26) Thiamine is useful in the treatment of peripheral neuropathies induced by which of the following?
A) arsenic and ethanol
B) ethanol and carbon dioxide
C) carbon dioxide and carbon disulfide
D) ethanol and carbon disulfide

(27) Ginger jake contaminated ginger extract was caused by
A) deltamethrin
B) methanol
C) triorthocresyl phosphate (TOCP)
D) triethyltin
E) acrylamide

(28) The most reliable animal model, to date, for the study of immunologic reactions of the delayed hypersensitivity type in contact photosensitivity dermatitis is the
A) rat
B) rabbit
C) guinea pig
D) gerbil
E) monkey

(29) Evidence for the allergic nature of photosensitization was obtained by
A) irradiation of active compounds before application to test animals
B) passive transfer of contact photoallergy from actively sensitized to non-sensitized guinea pigs
C) irradiation of test animals before application of active compounds
D) isolation of the photoantigen
E) correspondence of inducing wavelengths within absorption spectra of active compounds

Key for the following questions:
Choose the single most appropriate answer

(30) Which of the following is not a known skin sensitizer?
A) ethylenediamine
B) formaldehyde
C) TCDD
D) neomycin
E) 2,4-dinitrochlorobenzene

(31) The skin is an organ
A) incapable of metabolism
B) with little metabolic capacity
C) capable of substantial metabolism
D) whose metabolic capacity is fully known
E) none of the above

(32) Which of the following is/are true in regard to ocular toxicity?
A) the acute damage observed with an alkali burn is an accurate indicator of long term damage
B) the normal course of treatment with acid burns is prolonged irrigation with water even though with some acids such as H_2SO_4 considerable amounts of heat are generated which may damage the cornea
C) lime burns are like acid burns in that the material generates heat and tends to precipitate thereby making it difficult to remove from the eye
D) with alkali burns the use of steroids, prostaglandins and collagenase inhibitors are contraindicated
E) all of the above

(33) In which of the following conditions would the half saturation concentration for oxygen be lowest?
A) 50% decrease in hemoglobin
B) 25% carboxyhemoglobinemia
C) 50% methemoglobinemia
D) 50% carboxyhemoglobinemia
E) 25% methemoglobinemia

(34) Which of the following is characteristic of "altered foci" in rat liver?
A) decrease in gamma-glutamyl transpeptidase activity (↑)
B) increased glycogen storage
C) iron exclusion after iron loading
D) increase in glucose-6-phosphatase activity (↓)
E) always progress to neoplasia not true

(35) Increases in which of the following serum enzymes is most indicative of biliary tract damage rather than hepatocyte toxicity?
A) aspartate aminotransferase
B) alanine aminotransferase
C) lactic dehydrogenase
D) alkaline phosphatase
E) sorbitol dehydrogenase

(36) Lipofuscin accumulates in liver cells in chronic injury from carbon tetrachloride probably as a result of damage to
A) external cell membranes
B) mitochondria
C) lysosomes
D) endoplasmic reticulum
E) nuclear membrane

(37) Hepatic peroxisome proliferation in the rat has been produced by exposure to
A) clofibrate
B) diethylhexylphthalate
C) trichloroethylene
D) all of the above
E) none of the above

(38) Which of the following items is not characteristic of the phenomenon known as "light hydrocarbon nephropathy (LHN)"?
A) hyaline droplet formation in the renal proximal convoluted tubule cells
B) granular cast formation due to the sloughing of cellular debris into the renal tubular lumen
C) primarily observed following exposure to saturated aliphatic compounds that contain a ring in their structure
D) appears to be species and sex dependent male rats
E) postulated to involve a chemical interaction with alpha-2μ-globulin

Key for the following questions:
Choose the single most appropriate answer

(39) Damage to proximal convoluted tubules in the nephrons following exposure to inorganic mercury or uranium might be expected to be reflected in which of the following?

A) aminoaciduria
B) elevated aspartate aminotransferase (AST)
C) decreased glomerular filtration rate (GFR)
D) proteinuria
E) elevated blood glucose

(40) In order to properly select three dose levels for a subchronic toxicity study, which of the following factors is least important to know prior to the study

A) potential of the chemical to bioaccumulate
B) shape of the dose-response curve from acute studies
C) target organs
D) linearity of pharmacokinetics with dose

(41) The biotransformation of relatively inert chemicals to highly reactive metabolites (metabolic activation) is an important component of chemical-induced toxicities. It is true of this process that

A) phase I reactions (oxidations, reductions) generate reactive but protective metabolites
B) phase II reactions (conjugations) generate reactive toxic metabolites
C) a relatively inert chemical is usually harmless but once biotransformation occurs the end result is enhanced toxicity
D) all of the above
E) none of the above

(42) The initial oxidant species formed when an electron is transfered to molecular oxygen is

A) singlet oxygen
B) superoxide anion
C) hydroxyl radical
D) hydrogen peroxide
E) hydronium radical

(43) If the rate of excretion of a chemical is 0.3 Gm per hour, which of the following is true?

A) its biological half-life is 2.31 hours
B) it follows first order kinetics
C) it follows zero order kinetics
D) 30% of the chemical is excreted in an hour
E) none of the above

(44) A toxicant would be expected to accumulate in the body during a period of repeated exposure when the

A) volume of distribution is large
B) half-life is shorter than the exposure interval
C) half-life is longer than the exposure interval
D) toxicant is absorbed into soft tissues
E) rate of absorption is greater than the rate of excretion

(45) Which of the following microsomal inducers increase(s) bile flow and excretion?

A) benzo(a)pyrene
B) 3-methylcholanthrene
C) phenobarbital
D) all of the above
E) none of the above

(46) An example of the ability of genetic factors to influence the susceptibility to toxic agents is

A) genetic variation of glucuronyl transferase levels in cats
B) differences in the ability of dogs to acetylate arylamines
C) the sulfonamides' displacement of bilirubin from its plasma-binding sites in premature infants
D) hemolytic effects of primaquine in individuals with glucose-6-phosphate dehydrogenase deficiency
E) nephrotoxic effects of chloroform in mice

(47) Which of the following reactions is NOT inhibited by piperonyl butoxide?

A) aminopyrine N-demethylation
B) parathion desulfuration
C) aflatoxin B1 epoxidation
D) naphthalene hydroxylation
E) malathion hydrolysis

Key for the following questions:
Choose the single most appropriate answer

(48) The most common method used in extrapolating from animal subchronic toxicologic data to estimate an initial human human dose for a candidate drug is

A) based on low-dose risk estimation, e.g., Mantel-Bryan procedure
B) based on the safety factor approach using the type of toxicity, the comparative pharmacokinetic data and benefit consideration
C) based on the use of the body surface area
D) to divide the animal toxic dose by 10
E) none of the above

(49) There are four general phases to the risk assessment process. Initially, data are collected and evaluated and toxicology and exposure assessments are completed. The final stage of the risk assessment process integrates all of the above into the

A) integrated risk information system
B) health effects assessment summary
C) health effects advisory
D) risk characterization
E) health and environmental effects profile

(50) Short Term Exposure Limits (STELS) are

A) worker exposure limits for a single 15 minute interval during normal work cycle
B) promulgated by the United States Environmental Protection Agency
C) based on carefully controlled studies in humans
D) based on subchronic exposure studies in animals
E) none of the above

(51) Licensing, distribution and tolerance limits of pesticides are regulated by which governmental body?

A) Department of Agriculture
B) National Institute of Occupational Safety and Health (NIOSH)
C) Environmental Protection Agency (EPA)
D) Department of Transportation
E) none of the above

(52) Case fatality rate refers to

A) the crude death rate per 100,000 population
B) cause-specific death rate due to disease
C) a fatal outcome of any disease
D) the percentage of deaths among cases of the disease
E) the proportion of deaths due to a specific disease among all deaths from all causes

2 pmns

(53) Cohort studies are

A) designed to determine the effect of a specific chemical of interest on a population with a known exposure to the chemical
B) designed to determine the causative agent for a specific disease in a population afflicted with the disease
C) often useful in proving that a chemical is not a human carcinogen
D) a valuable tool for extrapolating animal data to humans
E) another name for case-control studies

(54) What percentage of a normally distributed population is represented by + or - 3 standard deviations about the mean?

SD

A) 33%
B) 50%
C) 68.3%
D) 95.5%
E) 99.7%

68.3 1
95.5 2
99.7 3

(55) Which of the following insecticides is NOT primarily a neurotoxicant?

A) DDT - OC
B) parathion - OP
C) pyrethrum - PY
D) dieldrin - OC
E) rotenone - resp distress

(56) In ecotoxicology, a sentinel species

A) is most abundant in the location of interest
B) is at the top of the food chain
C) contains a large, consistent amount of biomarkers
D) is exceptionally sensitive to a pollutant's effects
E) is opportunistic in its choice of habitats

Key for the following questions:
Choose the single most appropriate answer

(57) Which of the following air pollutants is not classified as a pulmonary irritant?
A) nitrogen dioxide
B) carbon monoxide
C) sulfur dioxide
D) formaldehyde

(58) Which of the following has NOT been implicated in causing metal fume fever?
A) aluminum
B) beryllium
C) copper
D) iron
E) tin

(59) All of the following statements concerning the biologic assessment of exposure are true EXCEPT
A) biologic monitoring of exposure includes an estimation of the internal dose
B) monitoring the airborne concentration of an industrial pollutant provides a reliable estimate of the amount absorbed by a worker
C) the concentration of polychlorinated biphenyls in blood is proportional to the amount accumulated in fatty tissue
D) biologic monitoring of exposure is of practical value only when relationships between external exposure, internal dose and adverse effects are known

(60) A wide spectrum of lung changes varying from a clinically insignificant focal pigmentation to crippling fibrosis is most likely to be encountered among which of the following?
A) distillery workers
B) coal miners
C) farmers
D) histologists
E) city traffic directors

(61) The use of British Anti-Lewisite (BAL) to treat poisoning by arsenic is an example of
A) functional antagonism
B) receptor antagonism
C) chemical antagonism
D) dispositional antagonism

(62) Renal papillary necrosis (RPN) has been associated with exposure to
A) mixtures of aspirin and phenacetin
B) aspirin
C) acetaminophen
D) nsaids (non-steroidal antiinflammatory drugs)
E) all of the above

(63) Opisthotonus, which may occur following a toxic dose of strychnine, refers to
A) paralysis of involuntary muscles
B) labored breathing
C) back arched in hyperextension with only the head and feet touching the ground
D) muscles of the face and neck relaxed
E) all of the above

Key for the following questions:
Choose A) if 1,2 and 3 are correct
Choose B) if 1 and 3 are correct
Choose C) if 2 and 4 are correct
Choose D) if only 4 is correct
Choose E) if all four answers are correct

(64) Heavy metal antagonists
1) react with heavy metals to form tightly bound complexes
2) react with heavy metal to form chelates
3) act by preventing the binding of toxic metals to body ligands
4) depend solely upon ring size of the resulting complex for chelate stability

(65) BAL (British anti-lewisite) is used in the treatment of
1) cadmium toxicity
2) selenium toxicity
3) tellurium toxicity
4) arsenic toxicity

(66) Benzene metabolism
1) does not lead to opening of the aromatic ring
2) results only in urinary phenol and catechol excretion
3) is affected by urinary pH
4) involves cytochrome P450, epoxide hydratase and glutathione S-epoxide transferase

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B

(67) The toxicity of methyl-n-butyl ketone (MBK) in humans is
1) expressed as a peripheral neuropathy
2) less than that of methyl ethyl ketone (MEK)
3) related to metabolites
4) has not been produced in experimental animals

More

(68) Dinitrophenols cause skin staining and can be very toxic if large quantities are absorbed. Immediate treatment to overcome toxicity includes
1) ice packs
2) atropine sulfate (slow - tachycardia)
3) remove material from skin
4) salicylates - Acid

B

Acetylcholine (SLUG)

(69) Insect growth regulators or juvenile hormones are a group of insecticides that
1) mimic compounds normally secreted by the corpus allatum in the insect brain
2) have target systems found in insects and not in mammals
3) control polymorphism in some insects
4) are neuropeptides maybe

A

(70) Which of the following suggest that lung damage induced by ozone may result from the oxidation of unsaturated fatty acids?
1) Mitochondria in alveolar macrophages swell after exposure to ozone
2) the ozone syndrome can be duplicated by the injection of the ozonide of linoleic acid
3) there is a greater penetration of serum proteins from blood into the air space after exposure of the test animals to ozone
4) alpha-tocopherol (vitamin E) prolongs the life of animals that are continuously exposed to a constant concentration of ozone in the inspired air

— ne Asue of tisperity

(71) A geometric standard deviation (sigma g) of less than 1.5 for particles within an aerosol means that the particles
1) are similar in size
2) will be deposited in the alveolar region
3) were probably generated in a laboratory
4) have a small aerodynamic diameter

B

(72) Which of the following are physical units for radioactivity?
1) dpm
2) Curie
3) Becquerel
4) rem — DOSE Biol unit

E

(73) Scombroid poisoning has occurred after the ingestion of
1) mackeral
2) mahi mahi
3) sardines
4) rainbow trout - fresh Ito Fish

SAH Ito Ash

A

(74) Hydrolysis of the glycoside amygdalin by enzymes in the gastrointestinal tract produces
1) benzaldehyde
2) glycyrrhizic acid
3) hydrocyanic acid
4) oxalic acid

Apricot Pits
Lgetrile

(75) Acetaminophen (paracetamol), when ingested at high doses, can cause hepatic necrosis. This tissue damage is caused by
1) oxidation of acetaminophen by a cytochrome P450-dependent monooxygenase system to an electrophilic metabolite that readily arylates tissue nucleophiles
2) oxidation of acetaminophen by NAD with alcohol dehydrogenase and catalysis to the aldehyde, which is hepatotoxic
3) oxidation of acetaminophen by a microsomal enzyme system to an electrophilic metabolite that combines with reduced glutathione (GSH) and, when the reduced GSH stores are depleted, initiates hepatotoxicity
4) dehydrogenation of the amino group by monoamine oxidase and subsequent dehydration to the aldehyde, which is hepatotoxic

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(76) Which of the following groups of drugs is/are correctly paired with the predominant toxic side effect?

1) beta adrenergic blocking agents (e.g., propranolol) - hypotension and bradycardia
2) classical antihistamines (e.g., diphenhydramine) - sedation
3) aminoglycoside antibiotics (e.g., gentamicin) - ototoxicity and renal toxicity
4) nonsteroidal anti-inflammatory drugs (e.g., ibuprofen) - gastrointestinal irritation

All E

(77) Cytological examinations are useful in the detection of the following types of mutations:

1) breaks
2) base-pair substitution
3) deletions
4) frameshift

Aberrations

(78) Test(s) to detect chromosomal mutations

1) Ames *salmonella* tests
2) dominant lethal test in mammals
3) specific locus test in mice
4) *in vivo* cytogenetic tests

(79) Which of the following are true of the dominant lethal assay?

1) both the male and female are exposed to the test chemical prior to mating
2) exposure of the female to the test chemical continues throughout gestation
3) dominant lethals are calculated as the ratio of dead-born fetuses to total fetuses
4) test designed to detect mutagenic effects in the spermatogenic process that are lethal to the embryo or fetus

mates only

(80) Which of the following is/are true integrative processes of the hypothalamo-pituitary-gonadal axis?

1) FSH and LH are glycoproteins synthesized and released from a subpopulation of the basophilic gondadotropic cells of the pituitary gland
2) luteinizing hormone-releasing hormone (LH-RH) acts on gonadotropic cells, thereby stimulating the release of FSH and LH
3) LH-RH and follicle-stimulating hormone-releasing hormone (FSH-RH) appear to be the same substance
4) gonadotropins are not important in the female

A

(81) It is currently claimed that primary lung cancer is

1) caused by heavy cigarette smoking over a period of years
2) decreasing in incidence in both males and females
3) related etiologically to air pollution
4) highly predictable on the basis of genetics

B

(82) The theoretical time for a test compound gas to reach 99% equilibrium in an inhalation chamber for a selected target concentration is termed the t99. In addition to the equation constant what measurements are needed to calculate the t99?

1) chamber volume
2) chamber target concentration of the test compound
3) chamber airflow rate
4) molecular weight of the test compound

B

(83) Peripheral axonopathies may be caused as a result of overexposure to which of the following agents?

1) acrylamide - gels
2) taxol - microtubules
3) ethyl alcohol -
4) 3,3'-iminodipropionitrile - IDPN - (TOCP)

E

waltzing syndrome

1,2,3 WMFT2
1,2,3

UV light

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(84) The blood-brain barrier is a functional concept for which the following can be stated:
1) highly polar compounds readily pass the barrier and enter the brain
2) the barrier is most likely due to the combination of a glial barrier and unique properties of brain capillaries
3) the barrier has been proven not to exist by definitive em studies
4) lipid-soluble compounds usually penetrate the barrier

C

(85) Which of the following tests are routinely used to measure the effect of immunotoxicants on B-cell activity?
1) plaque-forming cell response to sheep red blood cells spleen
2) complement depletion – Type II hypersensitivity
3) immunoglobulin quantitation
4) delayed hypersensitivity reactions – Type IV

BUSA humoral

B

(86) Which of the following metals can produce type IV allergic reactions?
1) nickel
2) chromium – 1° dermal
3) gold
4) beryllium – type-IV

B

(87) The rate of dermal absorption of toxicants may by facilitated by
1) increasing the toxicant's lipophilicity (for example, 2,4-D acid change to 2,4-D ester)
2) increasing skin blood flow
3) choice of solvent/vehicle
4) increasing skin water content

(88) Phototoxicity may result from
1) dermal exposure to furocoumarins, (e.g. 8-methoxypsoralen)
2) normal doses of certain antibiotics, (e.g. lomefloxacin or tetracycline)
3) exposure to petroleum products, (e.g. coal tar or pitch vapors)
4) dermal exposure to certain cationic detergents, (e.g. lauryl dimethyl benzyl ammonium) photosensitizer

A

(89) Eye irritation tests such as the draize rabbit test are not useful in evaluating damage to the
1) cornea
2) retina
3) conjunctiva
4) optic nerve

C

IRIS also
not Lens
PS 584

(90) Which of the following statements is/are true regarding surfactants and/or detergents and eye injury?
1) show a reasonable correlation between surface tension lowering capacity and damage produced
2) cationic agents are more damaging than anionic ones
3) are very likely to cause permanent opacity of the cornea
4) nonionic agents tend to be safer than either anionic or cation ones

C

(91) Signs of advanced chloroquine-induced retinopathy may include
1) progressive scotoma
2) narrowing of the retinal arteries
3) night blindness
4) abnormal electroretinograms

E

(92) Which of the following is/are true about methemoglobinemia?
1) methemoglobin is unable to bind oxygen
2) the oxidation of ferrous to ferric iron in one or more of the hemes affects the ability of other hemes to release oxygen
3) the red blood cell can reduce methemoglobin to hemoglobin
4) methemoglobin can form spontaneously

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(93) In nitrite poisoning
1) hemoglobin is oxidized to methemoglobin
2) the blood has a brown color
3) methemoglobin is ineffective as an oxygen carrier
4) methylene blue is an effective antidote because it is a reducing agent which rapidly converts methemoglobin to hemoglobin

(94) Increase the concentration of carbon monoxide in the blood by more than one percent
1) methylene chloride
2) smoking
3) hemolysis
4) automobile exhaust

(95) Which of the following statements concerning the kidney is/are true?
1) phenobarbital and salicylates will be cleared more rapidly from the plasma after IV administration of sodium bicarbonate
2) amphetamines will be cleared more rapidly from the plasma after IV administration of ammonium chloride
3) the mechanism for secretion of organic compounds is localized in the proximal tubule
4) because of high blood flow to the medulla, it is very sensitive to toxic insult

(96) The effect of toxic chemicals on renal function can be assessed by measurement of
1) inulin clearance
2) blood urea nitrogen
3) urine protein concentration
4) plasma creatinine

(97) Toxicity due to synergism between two drugs can most likely occur if the drugs
1) have similar pharmacologic activity
2) are toxic at high dose levels
3) have the same target organs
4) interact to mutually inhibit absorption across membranes

(98) Which of the following statements are characteristic for both resistance and tolerance?
1) effects are readily reversible
2) result from natural selection processes in the group
3) are due to genetic factors (inherited)
4) always result in a decrease in the toxic response

(99) Knowing that a drug is a base, pKa=10, and that its nonionized form has a low lipid solubility could enable one to predict that the drug would
1) not readily cross the placental barrier
2) have a small volume of distribution
3) penetrate the CNS poorly
4) have a long biological half-life ($t_{1/2}$)

(100) Which of the following agents have been useful in studying mechanisms associated with oxygen radical induced injury either *in vivo* or *in vitro*?
1) superoxide dismutase
2) hydroxyl radical scavengers
3) metal chelators
4) antioxidants

(101) The Monte Carlo procedure
1) is a statistical method used to distribute animals among dose groups and eliminate bias
2) results in a description of the probability distribution of risk
3) is most appropriate when group sizes are large, such as those found in carcinogenicity studies
4) is a statistical method which randomly selects values for each of many variables used as input for risk calculations

(102) Quantitaive cancer risk estimation, based on results of animal toxicity tests
1) is based on the results obtained at the maximum tolerated dose
2) is extrapolated only using linear dose-response relationships
3) is based on application of multiple safety factors
4) is based on both statistical procedures and mechanistic models of chemical carcinogenesis

Key for the following questions:
Choose A) if 1,2 and 3 are correct
Choose B) if 1 and 3 are correct
Choose C) if 2 and 4 are correct
Choose D) if only 4 is correct
Choose E) if all four answers are correct

(103) Which of the following is true about the federal OSHA Hazard Communication Standard (29CFR 1910.1200)?

EPA 1) it dictates labeling for all pesticides and consumer products used in industrial plants
2) it is known familiarly as the "worker right to know" law
EPA 3) it requires that mixtures which contain any detectable amount of a carcinogen must be labeled as carcinogens
4) it covers employees in the manufacturing sector

(104) The following test results are reportable to the EPA as required by section 8(e) of the Toxic Subtances Control Act

1) birth defects observed in an oral teratology study
2) an acute oral LD50 of 10 mg/kg
3) pronounced bioaccumulations in fish
4) a 20% decrease in organ weights in a subchronic feeding study – not much of a

(105) Epidemiological studies with immigrant populations are sometimes used to evaluate whether mortality rate differences between the populations are due to environmental factors or to a genetic composition. If observed differences are related to environmental differences, one would expect that

1) death rates for natives of the country of origin would equal the immigrant death rates
2) death rates for natives of the country of origin would differ from immigrant death rates
3) death rates for immigrants will differ from death rates for natives of the country of adoption
4) death rates for immigrants would equal death rates for natives of the country of adoption

(106) Organic matter in the soil have important absorptive properties for nonionic chemicals. The following are true – Acid

1) humic substances account for the majority of organic matter in soil – true
2) the "wick effect" increases soil absorption of nonionic chemicals – CATIONIC – not true
3) fresh or incompletely decomposed plant and animal material add to absorptive properties in soil – true
4) chemicals tend to become increasingly concentrated in the soil – degraded by microbes

(107) The carcinogenicity of the following chemicals has been demonstrated in animals. Which of the following have/has also been linked epidemiologically to cancer in humans after occupational exposure?

1) trichloroethylene – Solvent
2) styrene
3) chloroprene
4) vinyl chloride

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(108) The following statement(s) represent(s) the true principle(s) behind the treatment of cyanide poisoning with intravenous administrations of sodium nitrite and sodium thiosulfate

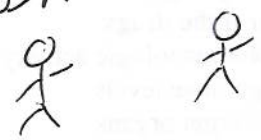
A

1) the affinity of methemoglobin for cyanide exceeds that of the cytochrome
2) sodium nitrite converts a fraction of circulating hemoglobin to methemoglobin
3) thiosulfate serves as a substrate for the enzyme rhodanese, which mediates the conversion of cyanide to thiocyanate
4) cyanomethemoglobin is an easily dissociable complex – stable

widely distributed
no need to induce

USA IMMIGRANT EU

USA EU



NO